



Reproducibility Report for ACM SIGMOD 2024 Paper: “Can Learned Indexes be Built Efficiently? A Deep Dive into Sampling Trade-offs”

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Abstract

In general, most of the main experiments in the paper “Can Learned Indexes be Built Efficiently? A Deep Dive into Sampling Trade-offs” can be successfully reproduced.

CCS Concepts

• Information systems → Data access methods.

Keywords

Learned Index

ACM Reference Format:

Yihao Liu, Yuanhui Luo, and Minguk Choi. 2024. Reproducibility Report for ACM SIGMOD 2024 Paper: “Can Learned Indexes be Built Efficiently? A Deep Dive into Sampling Trade-offs”. In *Reproducibility Reports of the 2024 International Conference on Management of Data (SIGMOD ARI Reports '24)*, June 9–15, 2024, Santiago, AA, Chile. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3687998.3717055>

1 Introduction

The paper “Can Learned Indexes be Built Efficiently? A Deep Dive into Sampling Trade-offs” [1] is written by MINGUK CHOI, SEE-HWAN YOO, JONGMOO CHOI from Dankook University. Overall, we think the authors did a good job trying to make the reproduction process easy.

2 Submission

The authors provide the following materials for the reproduction process:

- GitHub repository with code and scripts at: github.com/DKU-StarLab/BASIL
- A detailed compilation and setup instruction at the readme file.
- Detailed instructions for reproduction reviewers at github.com/DKU-StarLab/BASIL/blob/main/reproduce.md.
- A single reproduction bash script that includes all the experiment and figure plotting scripts.

3 Hardware and Software Environment

Below is the information about the environment used in the paper and in the reproduction process.

4 Reproducibility Evaluation

4.1 Process

Because the authors provide an “all-in-one” script, the reproduction is somehow easy. Although the compilation time is a little bit high, and several file paths in the scripts need to be manually modified at first. The authors have addressed and corrected the issues and suggestions raised during the review process, making the reproduction process simpler and more user-friendly. Generally, the authors did a great job of making the process easy.

4.2 Results

Here, present the key finding of the reproducibility evaluation. Ideally, all major findings of the paper can be reproduced. Note that not each and every figure of the paper needs to be reproduced for the paper to be reproducible. Rather, the figures that support the key results.

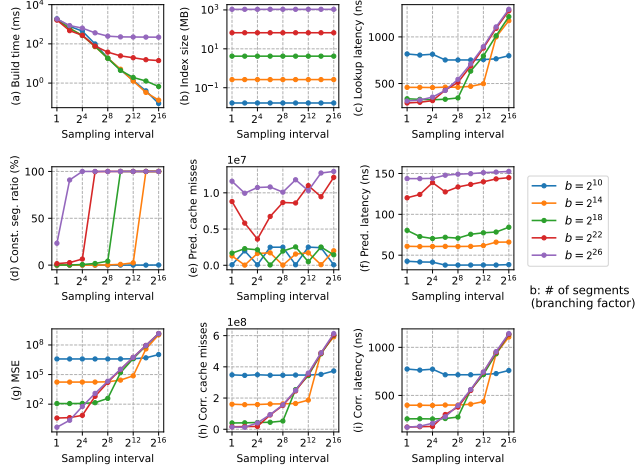
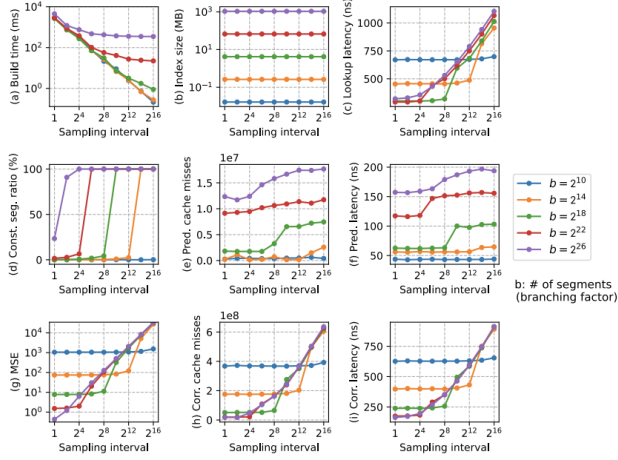
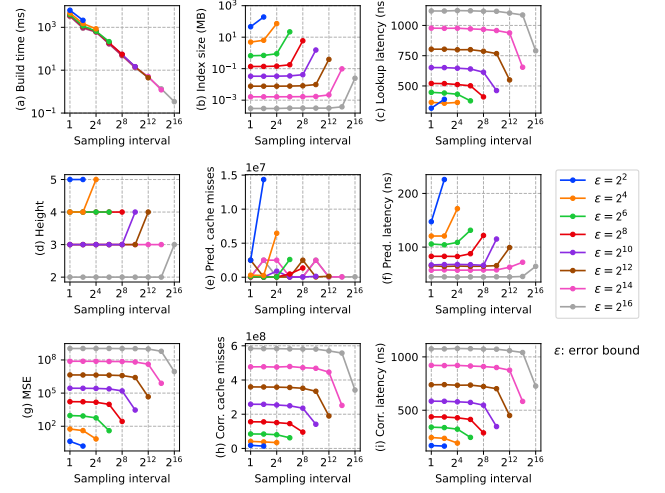
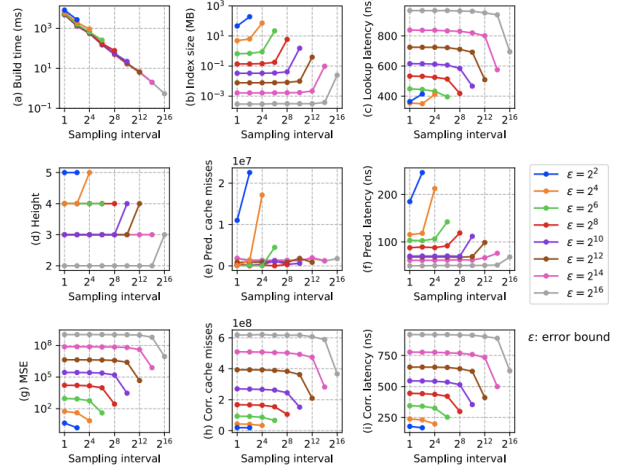
The primary key findings of the paper can be reproduced. Main results are as below.

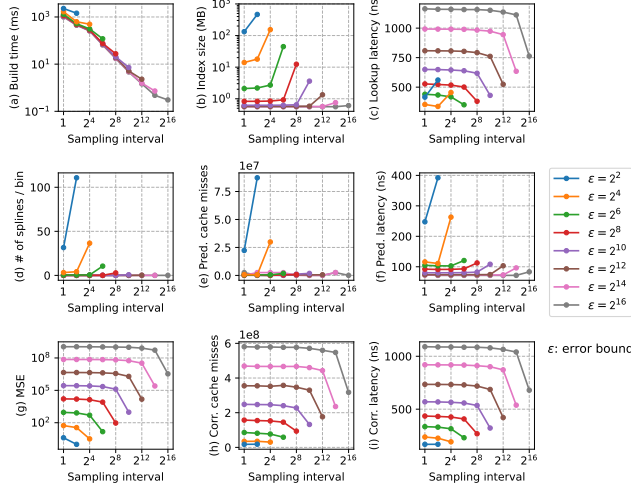


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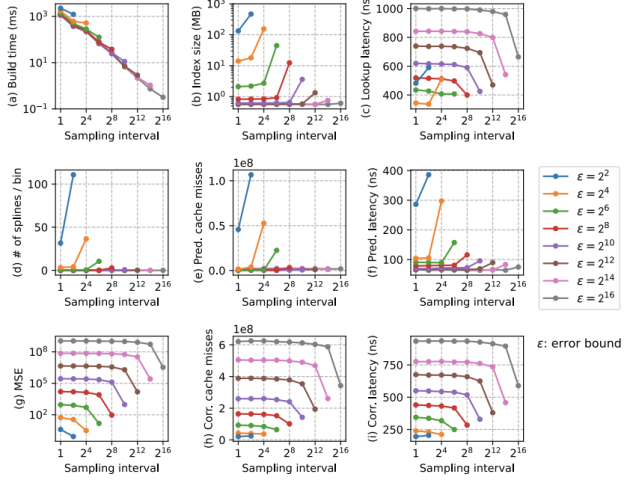
Table 1: Hardware & Software environment

	Paper	Repro Review #21B	Repro Review #21C
CPU	Intel(R) Xeon(R) Gold 6338	Intel(R) Xeon(R) Platinum 8474C	Intel(R) Xeon(R) Gold 5220
cores	32	96	18
GHz	2	2	2
RAM	512 GB	512 GB	192 GB
LLC	48 MB	97 MB	25 MB

**(a) reproduced version****(b) paper version****Figure 1: Figure 6: Performance impact of sampling on sRMI on history dataset.****(a) reproduced version****(b) paper version****Figure 2: Figure 7: Performance impact of sampling on sPGM on history dataset.**

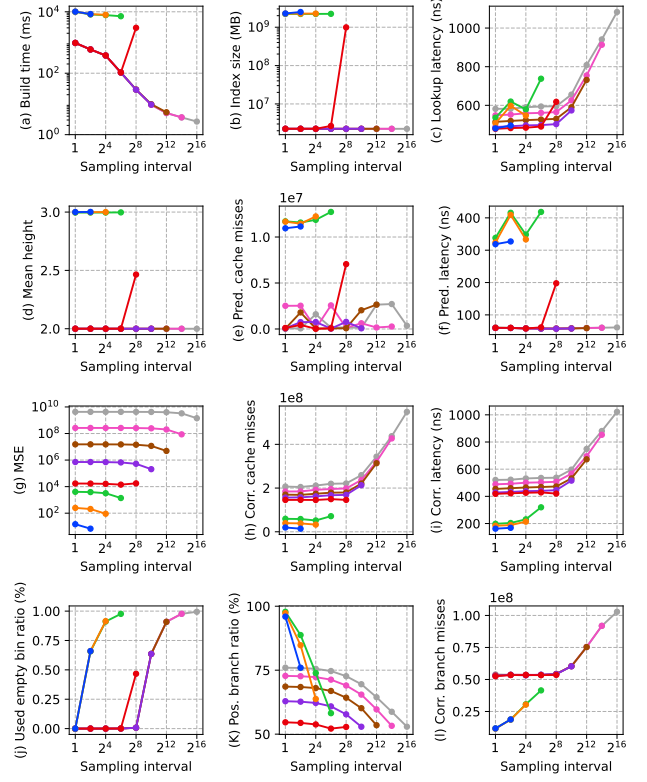


(a) reproduced version

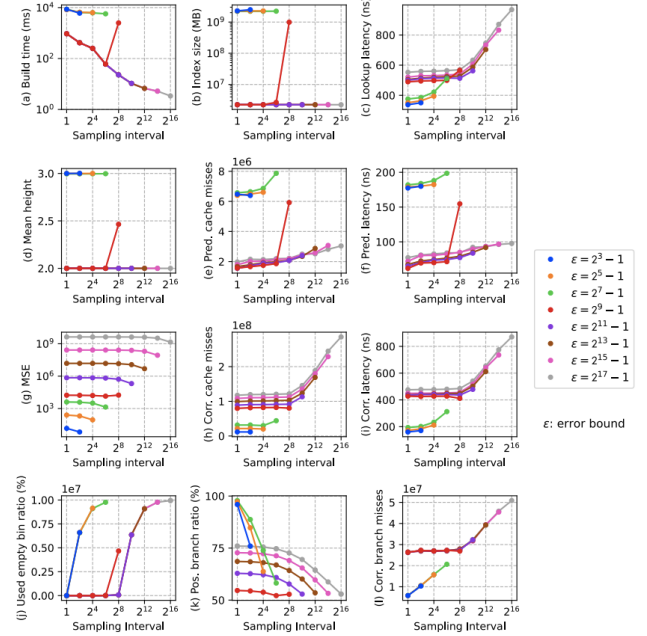


(b) paper version

Figure 3: Figure 8: Performance impact of sampling on sRS on history dataset.

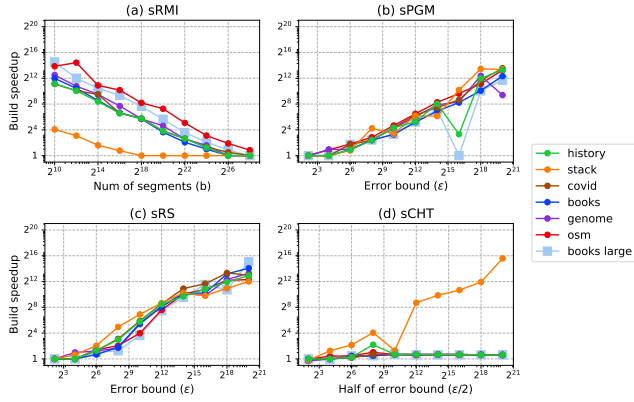


(a) reproduced version



(b) paper version

Figure 4: Figure 9: Performance impact of sampling on sCHT on history dataset.



(a) reproduced version

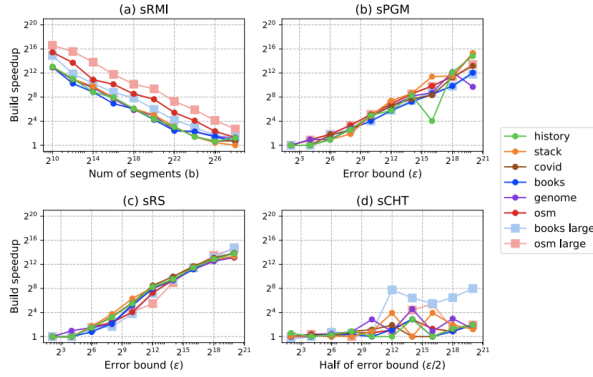
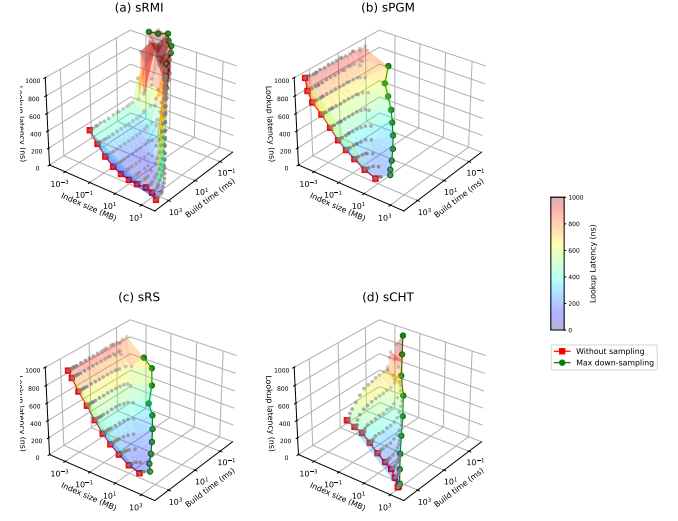


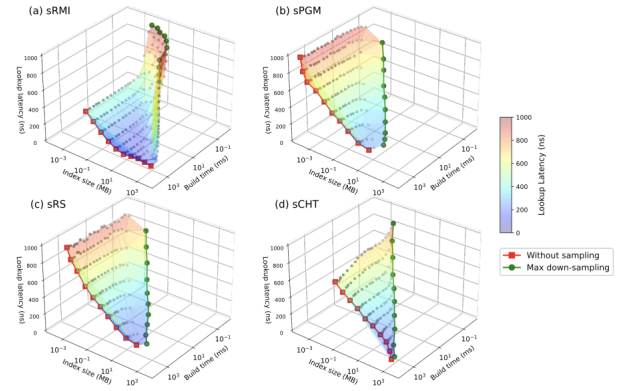
Fig. 10. build speedup with safe interval.

(b) paper version

Figure 5: Figure 10: build speedup with safe interval.



(a) reproduced version



(b) paper version

Figure 6: Figure 11: Design space of indexing schemes on history dataset.

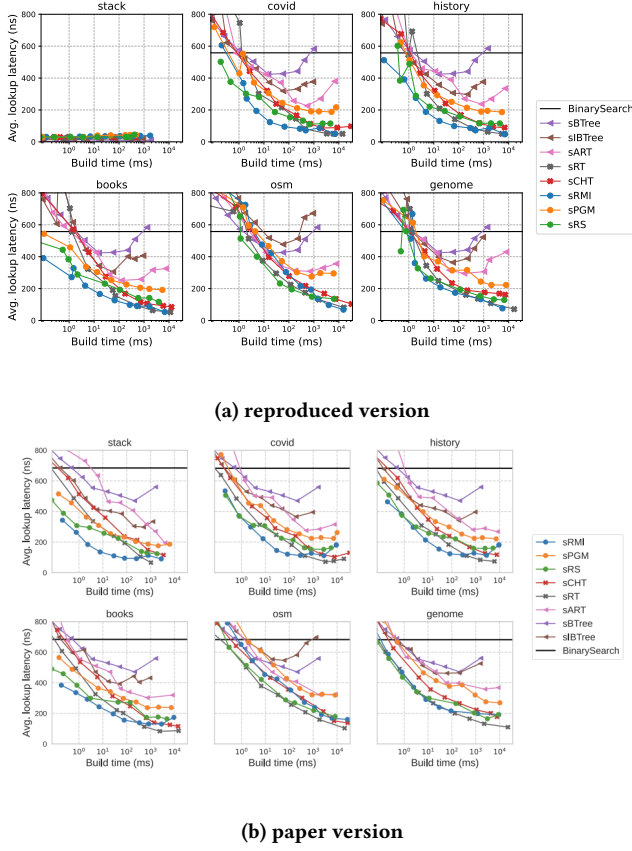


Figure 7: Figure 12: Average latency by the index build time.

	Instructions	Branch misses	TLB misses	LLC misses
Pred. Lat.	0.969	-0.209	0.938	0.985
correct	0.977	0.977	0.994	0.998

Figure 9: Table 2: Correlation coefficient between the performance counters with lookup latency.

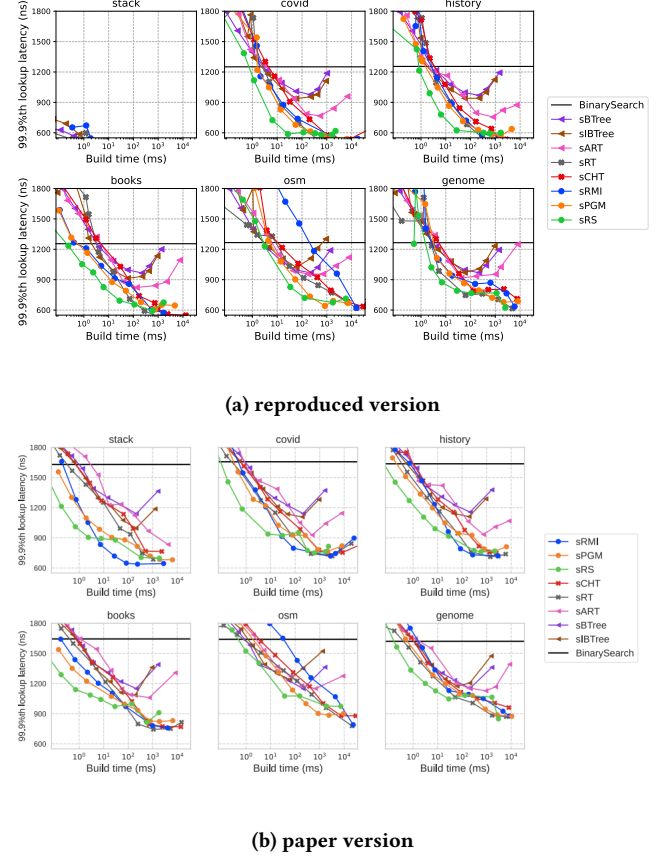


Figure 8: Figure 13: 99.9th percentile tail latency by the index build time.

5 Summary

As shown in the above figures, almost every core result and argument can be reproduced. Except for subgraph (d) of Figure 10 (in the original paper) and the results of the stack dataset in Figures 12 and 13 (in the original paper). In the reproduction process, several failures occur when operating on the stack dataset. Thus, the above misalignments can be related to these failures.

However, issues related to the stack dataset were observed by only one reviewer. In conclusion, the main experiments of this paper can be reproduced.

References

- [1] Minguk Choi, Seehwan Yoo, and Jongmoo Choi. 2024. Can Learned Indexes be Built Efficiently? A Deep Dive into Sampling Trade-offs. *Proceedings of the ACM on Management of Data* 2, 3 (2024), 1–25.